

Optical precursors in a weakly dispersive double narrow-resonance dielectric

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We investigate optical precursors transmitted through a double-Lorentz dielectric consisting of two adjacent narrow resonances. Compared to the patterns in a single-Lorentz dielectric, the transmitted envelope shows additional modulation that oscillates at the separation frequency between two resonances, corresponding to the doublet frequency separation. We use the weakly dispersive, narrow-resonance condition to analyze it, which compares favorably to our observations. This work will be meaningful to identify dielectric properties by analyzing multi-resonant media for application of optical communication. © 2019 Optical Society of America

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1. INTRODUCTION

Transient optical pulse propagation through a dispersive dielectric gives rise to pulse-envelope features known as optical precursors [1,2]. Precursors have received renewed interest because the so-called front velocity of the pulse is predicted to be precisely equal to the speed of light in vacuum c even when the pulse group velocity exceeds c [3–5]. Recent experiments have studied pulse propagation through a medium with narrow resonance when the optical carrier frequency is set to the resonance frequency, giving rise to nanosecond-scale precursors, and thus they are more readily observable with current laboratory equipment [6,7]. Besides their interest for classical pulse propagation, resonant precursors have been used effectively to engineer the quantum wavepacket of bi-photon states [8–10]. Motivated by these studies, optical precursors have also been observed in other materials with sharp resonances, including optically pumped crystals [11], four-wave mixing and light storage using cold atoms [12], inhomogeneously broadened media [13], and localized surface plasmons [14].

Resonant precursor formation has been studied theoretically using modern methods of coherent pulse propagation in a gas with a single narrow resonance [15–19], which assumes weak dispersion ($|\epsilon - 1| \ll 1$, where ϵ is the dielectric constant) and uses the rotating-wave approximation (RWA). These assumptions differ from the original work by Brillouin [1], who assumed that the optical carrier frequency is far from resonance and the medium has large dispersion in a regime where the RWA is likely not valid. In this case, asymptotic methods were used

to study pulse propagation, which have been further developed by Oughstun and Sherman [2] and colleagues [20]. To make a direct connection between these theoretical methods and thus demonstrate that the experiments truly observe precursors, Lefew *et al.* [21] applied asymptotic analysis to the weakly dispersive regime, $\omega_p \ll \omega_0 \delta$, where ω_p is a plasma frequency, ω_0 is a resonance frequency, and δ is a broadening of the resonance. Consequently, we found the analogy between the precursors and the coherent pulse propagation theory [22]. This work has been followed up by several other studies [6–10,21].

The primary focus of this paper is to extend these results to the case of a dielectric with two narrow and closely spaced resonances, as illustrated in Fig. 1(a) for the case when the optical pulse carrier frequency is scanned through the resonances. The transmitted field intensity starts with a full-amplitude peak at the beginning and then decays to the steady-state absorption level. For the off-resonance carrier frequency case, the transmission shows modulation depending on the detuning of carrier frequency from the medium resonance, as discussed in the single-resonance cases [7]. For the equally balanced double-resonance medium, as we present in this paper, a fast timescale modulation appears in addition to the slow timescale modulation observed in a single-resonance case. The analysis of optical precursors in a double-resonance medium is challenging because of the complexity and slowing convergence of a contour integral around the double resonances ω_1 and ω_2 .

In Brillouin's original work, he investigated optical precursor formation for a double-Lorentz medium (pp. 128–130 of

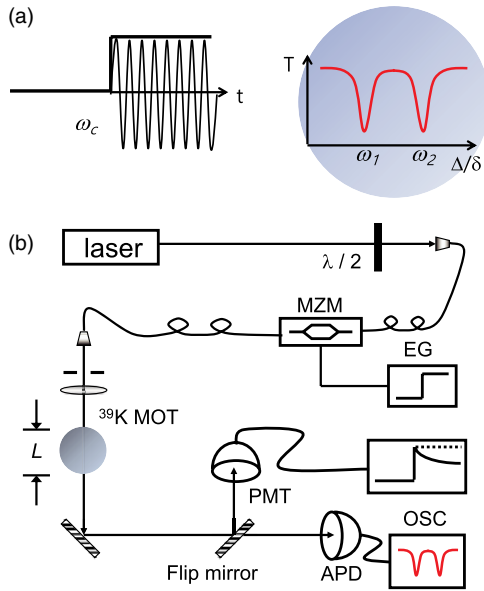


Fig. 1. Schematic diagrams of (a) step-modulated input pulse at the carrier frequency ω_c transmitted through a double-Lorentz dielectric, and (b) experimental setup. EG, edge generator; OSC, oscilloscope; APD, avalanche photodiode; PMT, photomultiplier tube; MZM, Mach-Zehnder modulator.

Ref. [1]). He discovered the existence of a third type of precursor (forerunner in his terminology) whose characteristics depend on the frequency difference ω_M between the two resonances. Oughtstun and Sherman [2] further developed the pulse-propagation theory for a highly dispersive double- or multi-resonance Lorentz dielectric and renamed the third type of forerunner a “middle precursor.” They found that the middle precursor arrives after the Sommerfeld precursor and is followed by the Brillouin precursor based on numerical analysis [2,23,24]. However, the middle precursor concept was developed for the highly dispersive broad absorption regime, where all precursor parts arrive separately. For the case of a weakly dispersive narrow-resonance (WDNR) dielectric, all transient parts arrive simultaneously with a rapid chirp [6,7,16]. In this regime, we show that it is possible to obtain analytic expressions that agree well with our experimental observations.

2. CHARACTERISTICS OF THE DOUBLE-LORENTZ DIELECTRIC

We consider a double-resonance dielectric medium realized by a gas of cooled and trapped ^{39}K atoms. Here, the carrier frequency ω_c of the optical pulse is set close to the $4S_{1/2}(F=1) \rightarrow 4P_{1/2}(F'=1, 2)$ transitions when the atoms are prepared in the $F=1$ hyperfine ground state via optical pumping [Fig. 2(a)], or the $4S_{1/2}(F=1) \rightarrow 4P_{1/2}(F'=1, 2)$ transitions when the atoms are pumped into the $F=2$ state [Fig. 2(b)]. In either case, the corresponding ground-to-excited-state transitions give rise to a double-Lorentz medium with a 58 MHz doublet separation, as seen in the right panels of Fig. 2, obtained by measuring the transmission of a weak continuous-wave laser passing through the gas. The strength of each resonance depends on the transition matrix elements

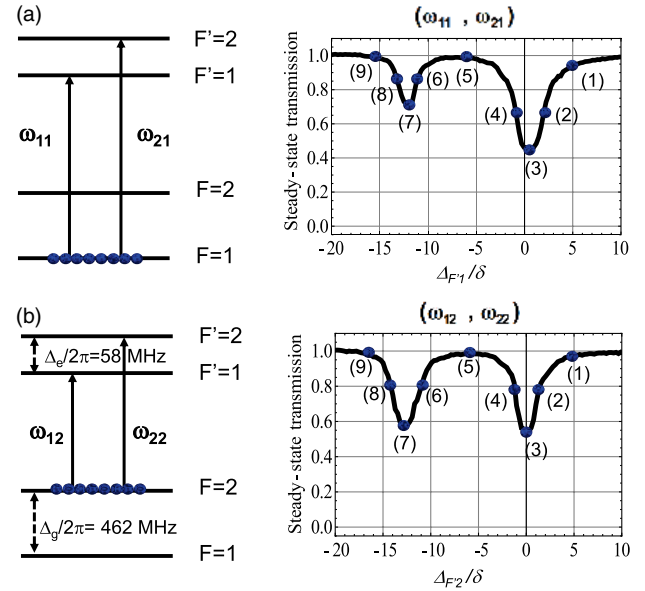


Fig. 2. (a) Energy-level diagram of the ^{39}K D_1 transition (left) and steady-state transmission as a function of $\Delta_{F'F}/\delta$ (right) when the atoms are pumped from the (a) $F=1$ and (b) $F=2$ ground hyperfine level into $F'=1, 2$ excited levels, respectively. We normalize the horizontal axis by $\delta = 2\pi \times 4.8$ MHz, as defined in the text. Here, $\omega_{FF'}$ indicates the various optical frequencies, $\Delta_g/2\pi$ ($\Delta_e/2\pi$) indicates the ground- (excited-) state splitting, and the blue dots, numbered (1)–(9), indicate the optical carrier frequencies where we perform pulse-propagation experiments.

and the optical pumping efficiency. For simplicity, we assume that the part of the optical spectrum overlapping with the other potassium resonances can be ignored. Frequency detunings from each resonance are defined through the relation $\Delta_{F'F} = \omega_c - \omega_{F'F}$. The resonance absorption linewidth [half-width at half-maximum (HWHM)] is denoted as δ in general. In this paper, we denote $\delta_{F'F}$ for four different transition cases, and $\delta \equiv \delta_{21}$.

3. EXPERIMENTAL PROCEDURES

To obtain a dielectric medium with a narrow double resonance experimentally, we build a magneto-optical trap (MOT) for cold potassium (^{39}K) atomic gas using laser beams tuned to the $4S_{1/2} \leftrightarrow 4P_{3/2}$ transition at 767 nm (D_2 transition) and anti-Helmholtz coils. The trapped atoms have a temperature of ~ 400 μK measured by the release-and-recapture method [25]. At such cold temperatures, the Lorentzian-like resonance linewidth is $2\delta/2\pi \sim 9.6$ MHz (full width at half-maximum), which is comparable to the 6-MHz natural linewidth. The estimated precursor time scale ($1/e$ decay) is $1/2\delta \sim 16.6$ ns [6]. By using an avalanche photodiode (APD, Thorlabs, Model APD430A), we determine the HWHM $\delta_{F'F}$ from the measurements of the frequency-dependent steady-state probe-laser-beam transmission at line center [Fig. 2] for each peak: $\delta_{11}/2\pi = 4$ MHz, $\delta_{21}/2\pi = \delta_{12}/2\pi = \delta_{22}/2\pi = 4.8$ MHz. We achieve an atomic number density of $1.2 \times 10^{10} \text{ cm}^{-3}$ by measuring the absorption of a weak continuous-wave probe beam passing through the MOT at the maximum intensities

of cooling and trapping laser beams ($\sim 33 \text{ mW/cm}^2$). The diameter of the MOT is determined by measuring the $1/e$ of the fluorescence as $L = 0.20 \text{ cm}$ (transmission path length) resulting in $T = 0.36$, corresponding to $\alpha_0 = 5.14 \text{ cm}^{-1}$, $\alpha_0 L = 1.03$, where the line-center absorption coefficient is $\alpha_0 \equiv \alpha_{21}$, determined through Beer's law $T = \exp[-\alpha_0 L]$.

We then optically pump the cold atoms into either of the ground states of the potassium $4S_{1/2}$ state, $F = 1$ or $F = 2$, thereby preparing a double-resonance Lorentz dielectric. To achieve optical pumping, the trap or repumping beam [red detuned from $4S_{1/2} (F = 1) \leftrightarrow 4P_{3/2}$ transition] is repeatedly switched off for $20 \text{ } \mu\text{s}$, and we conduct the optical precursor experiment. The trap or repumping beam is then turned back on for $80 \text{ } \mu\text{s}$, returning to MOT stage, and then the process is repeated. The optical pumping time interval of $20 \text{ } \mu\text{s}$ is chosen to reach equilibrium after optical pumping and provide enough time to perform the optical precursor experiment with a constant total number of atoms.

Once we optically pump the atoms, we send a weak-intensity, step-modulated pulse oscillating at the carrier frequency ω_c through the medium. Then we measure the intensity of the temporal evolution of transmitted pulse $T(z, t)$, where a weak continuous-wave laser at ω_c is modulated as a step-input pulse by passing it through a 20-GHz-bandwidth Mach-Zehnder modulator (MZM, EOSpace, Inc.) driven by an electronic edge generator (Data Dynamics, Model 5113). The edge is steepened using a back recovery diode (Stanford Research Systems, Model DG535), resulting in a rise time $\sim 100 - 200 \text{ ps}$. The peak intensity of the input pulse is $\sim 64 \text{ } \mu\text{W/cm}^2$, which is much less than the saturation intensity of the transition ($\sim 3 \text{ mW/cm}^2$) to avoid nonlinear optical phenomena. We use a fast-rise-time (0.78 ns) photomultiplier tube (PMT, Hamamatsu, Model H6780-20) to detect the transmitted pulse intensity. We measure the resulting electrical signal with a 1-GHz-analog-bandwidth digital oscilloscope (Tektronix, Model TDS 680B). The total edge rise time (10%–90%) is $\sim 1.7 \text{ ns}$, corresponding to a bandwidth of $\sim 206 \text{ MHz}$ for the complete system in the absence of the atoms. This time is short in comparison to the expected duration of the precursors.

The optical precursor experiment is conducted on the $4S_{1/2} \leftrightarrow 4P_{1/2}$ transition (D_1 transition, 770-nm transition wavelength), as shown in Fig. 2. There are two ground states ($F = 1, 2$) and two excited states ($F' = 1, 2$) at the D_1 transition. The corresponding four resonance peaks are denoted as $\omega_{F'F}$, where $F'F$ denotes the $4S_{1/2}(F) \leftrightarrow 4P_{1/2}(F')$ transition. To calibrate ω_c , we measure its detuning to the atomic resonance frequency via the steady-state transmission spectrum. From knowledge of the hyperfine splitting of the $4P_{1/2}$ state (58 MHz), we calibrate the horizontal axis of the scan. Then we determine the detuning $\Delta_{F'F} = \omega_c - \omega_{F'F}$ by comparing the steady-state transmission spectrum with the value of the long-time intensity of the transmitted pulse in our transient experiments with values $\Delta_{(1)} = 27.1^{+3.2}_{-2.1} \text{ MHz}$, $\Delta_{(2)} = 5.3^{+0.5}_{-0.4} \text{ MHz}$, $\Delta_{(3)} = 0^{+2.3}_{-2.3} \text{ MHz}$, $\Delta_{(4)} = -5.3^{+0.4}_{-0.5} \text{ MHz}$, $\Delta_{(5)} = -29.2^{+7.8}_{-6.5} \text{ MHz}$, $\Delta_{(6)} = -49.5^{+0.1}_{-0.1} \text{ MHz}$, $\Delta_{(7)} = -59.9^{+1.5}_{-1.4} \text{ MHz}$, $\Delta_{(8)} = -64.4^{+1.5}_{-2.0} \text{ MHz}$, and $\Delta_{(9)} = -82.8^{+3.4}_{-10.0} \text{ MHz}$, indicated in Fig. 3. Here, instead of using $\Delta_{F'F}$, we use $\Delta_{(i)}$, $i = 1, \dots, 9$ following the nine spectral points

in Fig. 2. The errors associated with these measurements arise from our mapping of transmission to the frequency and are asymmetric, as the slope of the curve is different at each point. The detuning $\Delta_{(i)}$ for each data set is chosen to minimize the error between the experimental data and the theory (χ^2 fit). The theoretically determined detuning parameters are almost identical to those measured in the experiment.

4. COMPARISON OF THE EXPERIMENTAL DATA TO THE FAST FOURIER TRANSFORM

We collect our experimental data from the two different double-resonance cases. One is for equal optical transition strength, and the other is for an asymmetric case. The transition of $4S_{1/2}(F = 1) \rightarrow 4P_{1/2}(F' = 1, 2)$ represents the asymmetric absorption strength, as shown in the black solid lines in Fig. 3, and the $4S_{1/2}(F = 2) \rightarrow 4P_{1/2}(F' = 1, 2)$ shows the equal transition strength in Fig. 4. We then compare our experimental data to the numerical fast Fourier transform (FFT, red dotted lines) for nine different carrier frequency detunings. We use the FFT to solve the transmitted electric field numerically. Comparing the data with the numerical FFT provides a reliable reference for two analytic approaches discussed later. For comparison with the experimental data, the numerical FFT results are convolved with the expression for a single-pole low-pass filter $dy(t)/dt = -\gamma_f(y(t) - T(z, t))$, where $y(t)$ is the filtered transmission function, and $\gamma_f = 2\pi$ (206 MHz) is the 3-dB roll-off frequency in rad/s. The filter reduces the transmission to $\sim 95\%$ immediately after the front. Figures 3 and 4 show our experimental data in the solid black lines, which agree well with the FFT predictions of the low-pass-filtered intensity transmission function in red dashed-dotted lines. The numerical FFT of $T(z, t)$ shows reasonable agreement with the experimental data.

Figure 3 shows transmitted intensities $T(z, t)$ at different carrier frequencies from the transition $4S_{1/2}(F = 1) \rightarrow 4P_{1/2}(F' = 1, 2)$. The notation of nine different detunings is the same as in Fig. 2(a). The off-resonance data taken at point (1), $\Delta_{21} = \Delta_{(1)} = 4.79\delta_{21}$, blue-detuned from ω_{21} , show steady-state transmission without absorption. The transient modulation patterns originate from the interference between the total precursors and the main signal [7]. Thus, the oscillatory period is inversely proportional to the carrier frequency detuning $\Delta_c \equiv \omega_c - \omega_{21}$. Consequently, the data taken at (2) for $\Delta_{21} = \Delta_{(2)} = 1.25\delta_{21}$ show longer modulation patterns followed by the steady-state transmission intensity level. Only the on-resonance transmission taken at (3) shows no modulation and decays exponentially with the time scale of $1/2\delta_{21} \sim 16.6 \text{ ns}$. As we go to the red-detuned off-resonance frequencies at points (4) and (5), the patterns look similar to the case of (2) and (1), which obey the single-resonance approximation [6,7].

On the other hand, if we detune further to point (6–9) taken near the weak absorption peak ω_{11} , we observe a distinct modulation pattern of 17 ns period, the faster modulation compared to the detuning $\Delta_{F'F}$ [7]. As we will discuss theoretically later, this timescale corresponds to the excited state splitting $\Delta_c/2\pi = 58 \text{ MHz}$. Therefore, the fixed fast modulation pattern in the data might serve as a reference to test whether a dielectric medium can be treated as a single-resonance or

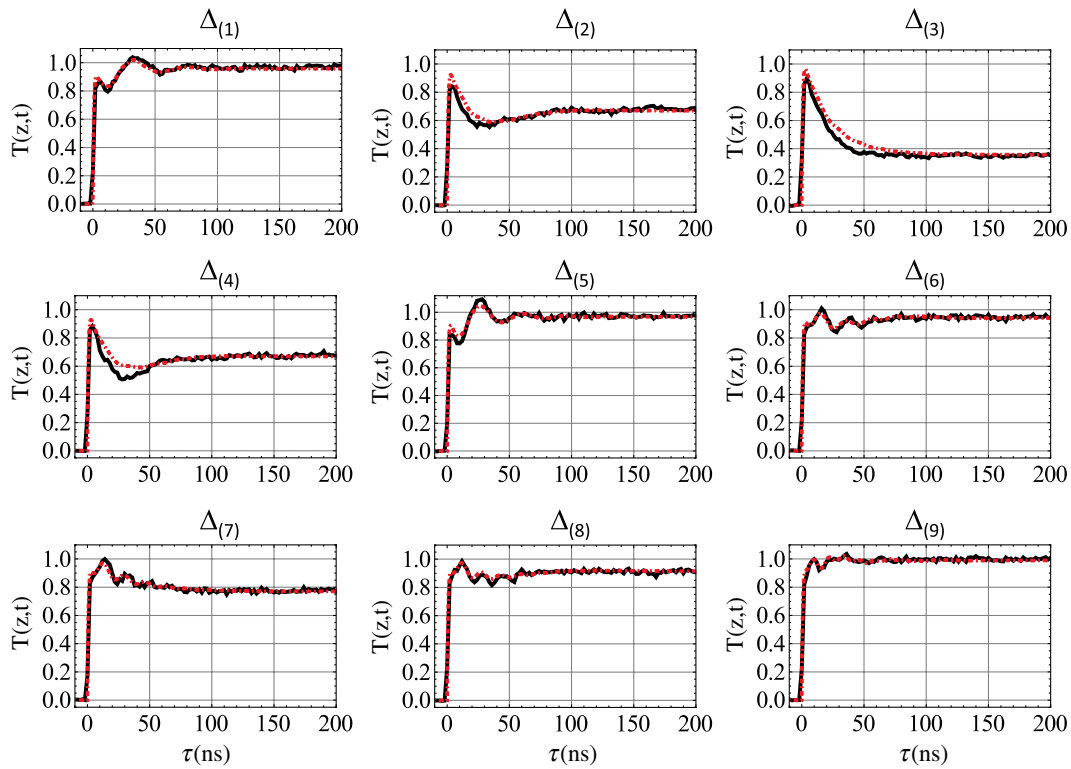


Fig. 3. Temporal evolution of transmitted intensity through the cloud of atoms. Solid lines denote experimental data taken at nine different frequencies near $4S_{1/2}(F=1) \rightarrow 4P_{1/2}(F'=1, 2)$ transition indicated in Fig. 2(a). Red dots are predicted by numerical FFT.

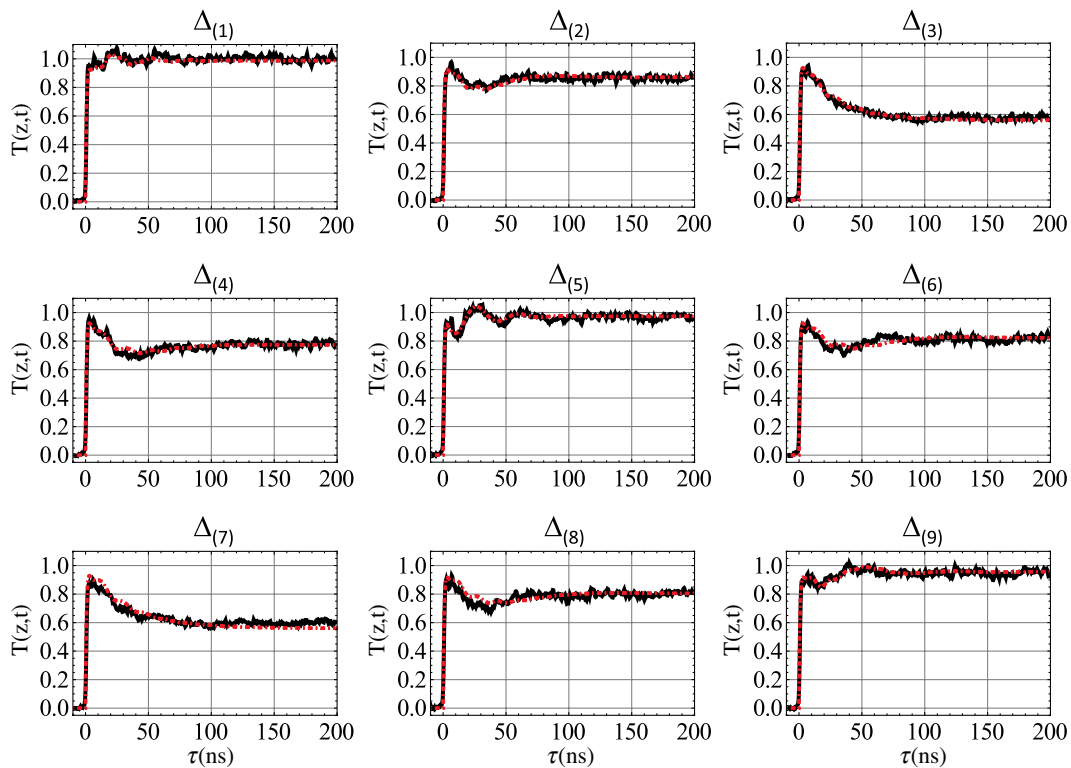


Fig. 4. Temporal evolution of transmitted intensity through the cloud of atoms. Solid lines denote experimental data taken at nine different frequencies near $4S_{1/2}(F=2) \rightarrow 4P_{1/2}(F'=1, 2)$ transition indicated in Fig. 2(b). Red dots are predicted by numerical FFT.

double-resonance system. For example, data set (3) has mild modulation, while data set (7) has strong modulation. This is because the relatively strong absorption peak at ω_{21} affects the weak one, while the weak one barely affects the strong one. Therefore, this would be evidence that the strong absorption line near ω_{21} can be treated as a single-resonance model, as in Refs. [6,7].

Figure 4 shows the total transmission for the other pair of optical transitions, $4S_{1/2}(F=2) \rightarrow 4P_{1/2}(F'=1, 2)$, demonstrating equally balanced double resonance, as illustrated in Fig. 2(b). We observe the fast oscillatory patterns associated with the $\Delta_e/2\pi = 58$ MHz for all nine cases contrary to the asymmetric case in Fig. 3, although the modulation intensities are small. The plots at (1) and (5) and the last one at (9) do not differ from the input step modulated intensity except for the initial oscillatory behavior from the off-resonance carrier frequency. As the carrier frequencies move close to the resonance, the $\Delta_{F'F}/2\pi$ -dependent oscillation becomes longer, as we see in data (2), (4), (6), and (8). The on-resonance cases of (3) and (7) show only exponential decay with timescale $1/2\delta_{F'F}$.

5. ASYMPTOTIC ANALYSIS

To establish the theoretical model of optical precursors for the case of a WDNR double-Lorentz dielectric, we assume that the precursor fields are simply the superposition of a precursor field arising from each single-resonance line. For the main signal, we apply the absorption coefficient $\alpha_{F'F}(\omega)$ for the double-resonance case from $n(\omega)$ instead of the single-resonance case:

$$n(\omega) = \sqrt{\epsilon(\omega)} = \sqrt{1 - \frac{\omega_p^2}{\omega^2 - \omega_0^2 + 2i\omega\delta}}, \quad (1)$$

where $\epsilon(\omega)$ is the linear frequency-dependent dielectric constant, ω_p is the plasma frequency, δ is the resonance absorption linewidth (HWHM), and ω_0 is the atomic resonance frequency. By considering our experimental condition of a WDNR dielectric medium, $\omega_{F'F} \gg \omega_{p_{F'F}}$, $\delta_{F'F}$ and $\omega_{p_{F'F}}^2 \ll \omega_{F'F}\delta_{F'F}$, we obtain a simplified expression of $n(\omega)$ for the case of double resonance:

$$n(\omega) \simeq \sum_{F'F} \left(1 - \frac{1}{4} \frac{\omega_{p_{F'F}}^2}{\omega(\omega - \omega_{F'F} + i\delta_{F'F})} \right), \quad (2)$$

where F [F'] represents the ground [excited] state hyperfine level: $4S_{1/2}(F=2) \rightarrow 4P_{1/2}(F'=1, 2)$ or $4S_{1/2}(F=1) \rightarrow 4P_{1/2}(F'=1, 2)$. The analytic expression of a main signal can be obtained by the contour integral around ω_c . Unfortunately, it is impossible to get an analytic solution for the total precursor field even with the approximations that lead to Eq. (2).

The field amplitude, $E(z, t)$, in terms of its envelope, $A(z, t)$, is expressed as

$$\begin{aligned} E(z, t) &= A(z, t)e^{-i\omega_c\tau}, \\ E_{S[B]}(z, t) &= A_{S[B]}(z, t)e^{-i\omega_{S[B]}\tau} = A_{S[B]}(z, t)e^{-i\omega_0\tau}, \\ E_C(z, t) &= A_C(z, t)e^{-i\omega_0\tau} = A_C(z, t)e^{i\Delta\tau}e^{-i\omega_c\tau}, \end{aligned} \quad (3)$$

where ω_c is a carrier frequency of the input pulse, and $\tau \equiv t - z/c$ is retarded time. Subscripts S , B , and C denote the Sommerfeld precursor, Brillouin precursor, and main signal, respectively. The frequencies of Sommerfeld and Brillouin precursors experience rapid chirp [6], so that these frequencies reach the medium characteristic resonance frequency ω_0 . In general, each part of precursors $A_S(z, t)$, $A_B(z, t)$, and $A_C(z, t)$ can be evaluated by using asymptotic analysis from Sommerfeld and Brillouin [1], and the advanced asymptotic analysis by Sherman and Oughstun [2]. To understand the behavior of the double resonance in this regime, we simplify the general analysis for a single-resonance case first, and then apply our postulate of linear superposition. Then the simplified expression of each part of Sommerfeld and Brillouin precursors of the double resonance is given as

$$\begin{aligned} \tilde{A}_{S_{F'F}}^{\text{asymp}}(z, t) &= - \sum_{F'=1,2} \frac{\sqrt{3}}{\sqrt{3}-1} A_0 \Theta(\tau) s(z, \Delta_{F'F}) \\ &\times e^{-\delta_{F'F}\tau - i\Delta_{F'F}\tau} e^{i\phi_{SB_{F'F}}}, \end{aligned} \quad (4)$$

$$\begin{aligned} \tilde{A}_{B_{F'F}}^{\text{asymp}}(z, t) &= \sum_{F'=1,2} \frac{1}{1-\sqrt{3}} A_0 \Theta(\tau) s(z, \Delta_{F'F}) \\ &\times e^{-\delta_{F'F}\tau - i\Delta_{F'F}\tau} e^{i\phi_{SB_{F'F}}}, \end{aligned} \quad (5)$$

$$\begin{aligned} \tilde{A}_{SB_{F'F}}^{\text{asymp}}(z, t) &= \sum_{F'=1,2} A_0 \Theta(\tau) s(z, \Delta_{F'F}) \\ &\times e^{-\delta_{F'F}\tau - i\Delta_{F'F}\tau} e^{i\phi_{SB_{F'F}}}, \end{aligned} \quad (6)$$

$$s(z, \Delta_{F'F}) = \sqrt{1 + e^{-\frac{\alpha_{F'F}z\delta_{F'F}^2}{(\Delta_{F'F}^2 + \delta_{F'F}^2)}} - e^{-\frac{\alpha_{F'F}z\delta_{F'F}^2}{2(\Delta_{F'F}^2 + \delta_{F'F}^2)}} \cos \left[\frac{\Delta_{F'F}\alpha_{F'F}z\delta_{F'F}}{2(\Delta_{F'F}^2 + \delta_{F'F}^2)} \right]}, \quad (7)$$

$$\phi_{SB_{F'F}} = -\text{Arg}[1 - e^{\alpha_{F'F}z\delta_{F'F}/[2(i\Delta_{F'F} - \delta_{F'F})]}], \quad (8)$$

where $\Delta_{F'=1,F} - \Delta_{F'=2,F} = \Delta_e$. As discussed in Ref. [6], the gap between the original theory and our experimental regime can be overcome by the scale factor $s(z, \Delta_{F'F})$. After the rapid chirp, the two types of saddle points coalesce, and $\tilde{A}_{S_{F'F}}^{\text{asymp}}(z, t)$ and $\tilde{A}_{B_{F'F}}^{\text{asymp}}(z, t)$ become identical except for the phase and amplitude [16]. When we consider the explicit form of the double-resonance case, we obtain an expression demonstrating the modulation at Δ_e because of the relation to the transmitted intensity $T(z, t) = |A(z, t)|^2$ as

$$\tilde{A}_{SB}^{\text{asymp}}(z, t) = A_0 \Theta(\tau) s(z, \Delta) e^{-\delta\tau - i\Delta\tau} e^{i\phi_{SB}} (1 + e^{-i\Delta_e\tau}). \quad (9)$$

On the other hand, the envelope of the main signal is obtained from the steady-state response (integration around the pole at ω_c) [2] and given as

$$\tilde{A}_C^{\text{asymp}}(z, t) = 2\pi i \text{Res}(\omega = \omega_c) \sim A_0 \Theta(\tau) e^{\sum_{F'=1,2} \frac{\alpha_{F'F}z\delta_{F'F}/2}{i\Delta_{F'} - \delta_{F'F}}}. \quad (10)$$

6. ROTATING WAVE APPROXIMATION

In the WDNR regime, and using the RWA, we can obtain an analytic analysis without a scale factor, as discussed in the single-resonance case [6]. Contrary to the previous section, however, this full analytic method cannot provide each of the Sommerfeld and Brillouin precursor parts separately. Nonetheless, we also show the interference of two single-resonance optical precursors indicating the presence of a double resonance.

We apply the same postulate as in the previous section: the precursor fields are the superposition of a precursor field arising from each single-resonance line. The main signal has a change in the absorption coefficient $\alpha_{F'F}(\omega)$ only for the double-resonance case in comparison to the single-resonance case. The total precursor and the main signal envelopes are given as

$$\tilde{A}_{SB}(z, t) = -A_0 \Theta(\tau) \sum_{F'=1,2} \left\{ e^{(i\Delta_{F'F} - \delta_{F'F})\tau} \times \sum_{n=1}^{\infty} \left(\frac{p_{F'F}}{i\Delta_{F'F} - \delta_{F'F}} \right)^n (p_{F'F})^{-\frac{n}{2}} J_n(2\sqrt{p_{F'F}}) \right\}, \quad (11)$$

$$\tilde{A}_C(z, t) = A_0 \Theta(\tau) e^{\sum_{F'=1,2} \frac{p_{F'F}}{i\Delta_{F'F} - \delta_{F'F}}}, \quad (12)$$

where $p_{F'F} \equiv \alpha_{F'F} z \delta_{F'F} / 2 = \omega_{p_{F'F}}^2 z / 4c$. The amplitude in Eq. (11) is also proportional to $(1 + e^{-i\Delta_e \tau})$, and consequently the intensity $T(z, t) = |A(z, t)|^2$ shows oscillation at $\Delta_e / 2\pi$, explaining the rapid modulation appearing in our experimental data. The RWA analysis is compared to the data sets in the blue

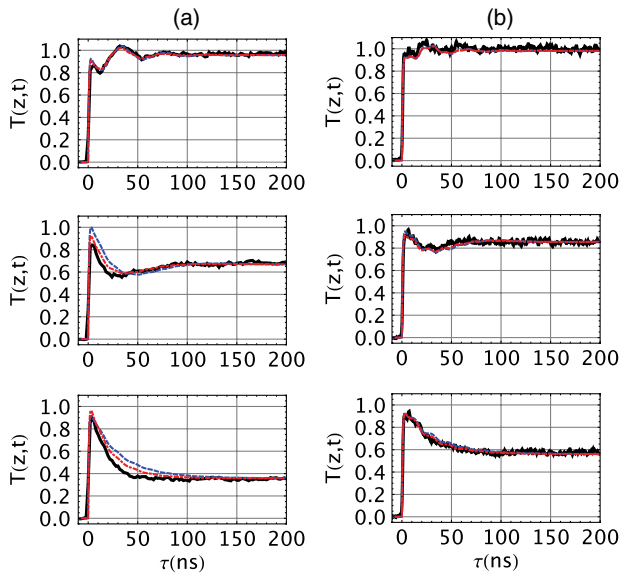


Fig. 5. Comparison of the theories. Experimentally obtained transient transmission intensity (black solid lines) compared with two theoretical analyses: the asymptotic analysis [Eqs. (4)–(10), red dashed-dotted lines], and the weakly dispersive narrow resonance [Eqs. (11), (12), blue dashed lines]. Transient transmission taken near the (a) $4S_{1/2}(F=1) \leftrightarrow 4P_{1/2}(F'=1, 2)$ transition, and (b) $4S_{1/2}(F=2) \leftrightarrow 4P_{1/2}(F'=1, 2)$ transition for $\Delta_{F'F} = \Delta_{(1)} \sim 5\delta$ (top plots), $\Delta_{F'F} = \Delta_{(2)} \sim \delta$ (middle plots), and $\Delta_{F'F} = \Delta_{(3)} \sim 0$ (bottom plots).

dashed lines in Fig. 5. The theoretically predicted envelopes of each field [Eqs. (11) and (12)] are shown in Figs. 6(b) and 6(d) for points (1)–(3).

7. DISCUSSION

With two theoretical approaches discussed in the previous sections, we investigate our data [Figs. 3 and 4] in this section. Figure 5 shows comparison among experimental data (black solid lines), asymptotic analysis (red dashed-dotted lines), and RWA (blue dashed lines). Note that we apply a single-pole low-pass filter $dy(t)/dt = -\gamma_f(y(t) - T(z, t))$ to the theories to compare them with the experimental data. The total transmitted field envelopes $T(z, t)$ evaluated from the asymptotic theory of the double-resonance model [Eqs. (4)–(10)] are compared to the experimental data taken near the $4S_{1/2}(F=1) \leftrightarrow 4P_{1/2}(F'=1, 2)$ transition for the case of spectral point ID (1)–(3) (red dashed dots) in Fig. 5(a). The $4S_{1/2}(F=1) \leftrightarrow 4P_{1/2}(F'=1, 2)$ transition has a pair of resonance peaks at $\omega_{F'F} = (\omega_{11}, \omega_{21})$, as shown in Fig. 2(a). Each resonance has a different absorption coefficient so that the one peak is weak and the other is strong. Figures 6(a)–6(b) show the envelopes of total precursors $|A_{SB}(z, t)/A_0|$ (first row), the main signal $|A_C(z, t)/A_0|$ (second row), and the total transmitted field $T(z, t)$ for the case in Fig. 2(a). For the equally balanced transition, $4S_{1/2}(F=2) \leftrightarrow 4P_{1/2}(F'=1, 2)$ [Fig. 2(b)], Fig. 5(b) shows the total transmitted field envelopes $T(z, t)$ compared to both asymptotic and RWA analysis. Figures 6(c)–6(d) show only the theoretical analysis: envelopes of total precursors $|A_{SB}(z, t)/A_0|$ (first row), the main signal $|A_C(z, t)/A_0|$ (second row), and the total transmitted field $T(z, t)$.

The first row in Fig. 6 shows the envelope of the total precursor, given in the asymptotic expression of Eq. (6) and RWA of Eq. (11), for different carrier frequency detunings $\Delta_{F'F} = \omega_c - \omega_{F'F} = 0, 1.25\delta_{F'F}, 4.79\delta_{F'F}$. The timescale of the modulation on the envelope does not depend on $\Delta_{F'F}$. Instead, it is fixed to the frequency separation between the two resonances arising from the excited-state hyperfine splitting $\Delta_e \equiv \omega_{21} - \omega_{11} = \omega_{22} - \omega_{12} \sim 12\delta_{F'F}$ comparable to the rapid oscillation on the transmitted intensity. Therefore, the fast oscillation with a period $2\pi/\Delta_e$ indicates the double-resonance effect in the experimental data.

Figures 6(a) and 6(c) show the asymptotic analysis [Eqs. (6)–(10)] compared to the RWA case in Figs. 6(b) and 6(d) [Eqs. (11) and (12)] demonstrating agreement between the two approaches. For example, the fixed modulation of the precursor envelopes in Figs. 6(a) and 6(b) is rather mild compared to that of the other pair shown in Figs. 6(c) and 6(d). This implies that the $4S_{1/2}(F=1) \leftrightarrow 4P_{1/2}(F'=1, 2)$ transition is primarily single resonance, as discussed in the previous sections, supported by the transmitted intensity data shown in Fig. 3. There, data set (3) barely shows the fixed modulation due to the separation of the two resonances, while data set (7) shows distinct modulation patterns. In this regard, the other pair is treated as a double resonance because of the modulation at both (3) and (7). For the pair near $4S_{1/2}(F=2) \leftrightarrow 4P_{1/2}(F'=1, 2)$ transition, the fast modulation in Figs. 6(c)–6(d) appears throughout all nine data sets in Fig. 4. This indicates the double-resonance

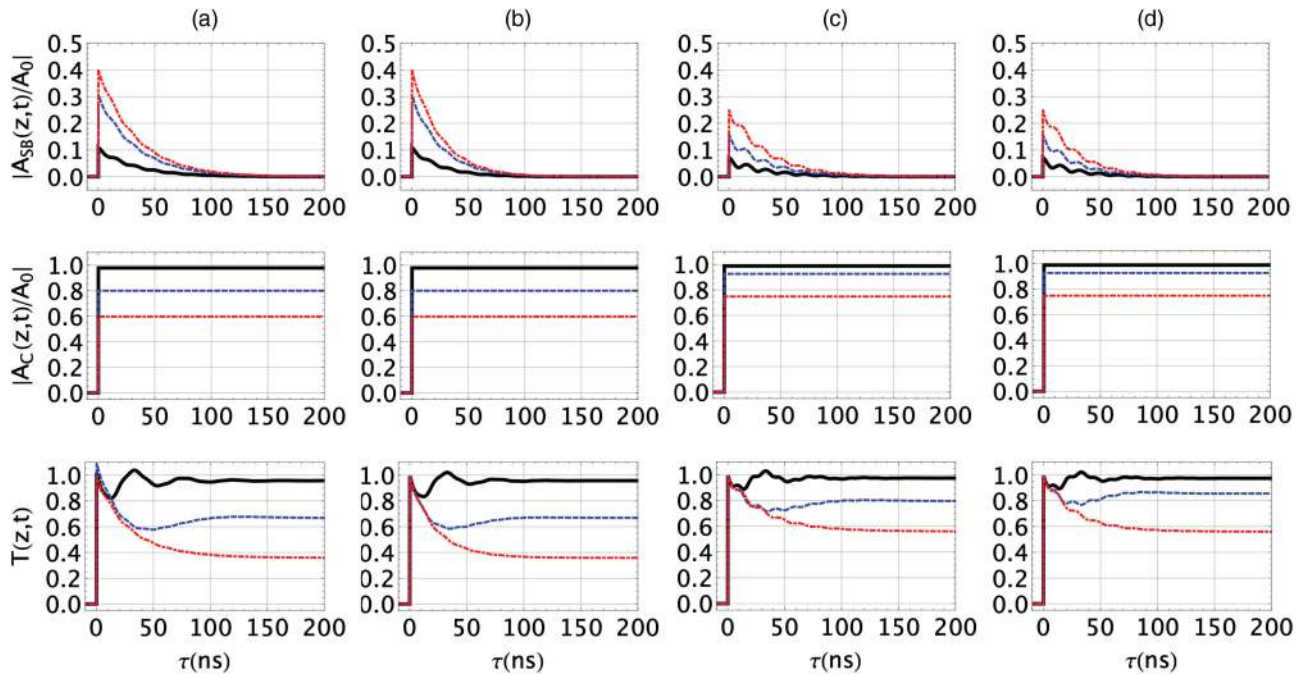


Fig. 6. Comparison of the theories between asymptotic [(a), (c)] and RWA [(b), (d)]. From the top row, the envelope of the total precursor, main signal, and total transmission are plotted for a double resonance for the case of $4S_{1/2}(F=1) \leftrightarrow 4P_{1/2}(F'=1, 2)$ [(a), (b)] and $4S_{1/2}(F=2) \leftrightarrow 4P_{1/2}(F'=1, 2)$ [(c), (d)]. All plots show three different detuning cases: $\Delta_{F'/F} = \Delta_{(3)} = 0$ (red dashed dots), $\Delta_{F'/F} = \Delta_{(2)} = 1.25\delta$ (blue dotted lines), and $\Delta_{F'/F} = \Delta_{(1)} = 4.79\delta$ (black solid lines).

pair at $(\omega_{12}, \omega_{22})$ needs to be treated as a double-resonance model, while the relatively strong peak in the other pair $(\omega_{11}, \omega_{21})$ can be regarded as single resonance.

The range of validity of the WDNR regime discussed in this paper has parameter sets $(\omega_c, \omega_p, \text{ and } \delta)$ different from those in Refs. [2,23,26] for the case of a highly dispersive (mature) double-resonance dielectric. In the future, it would be interesting to use advanced theoretical methods for understanding pulse propagation through a double-resonance material that spans both the weak and mature dispersion regimes. The general analysis will require one to overcome the issues arising from the complicated double-resonance analysis. Then, we may understand the behavior of the pulse evolution near the intermediate values between the mature (highly dispersive) and immature (weakly dispersive) dielectric in detail. From the advanced analysis, we may double-check whether the oscillatory responses arise from the immature medium characteristics [26]. We can also investigate the signal or energy velocity behavior, as discussed in the mature dielectric [2,23,26].

In summary, for the case of well-separated resonances (but close enough to affect each other, especially for the balanced double resonance) we find that the transmitted field consists of the Sommerfeld and Brillouin precursor pairs arising from both resonances. The main signal, however, is obtained by the absorption coefficient of the double resonance. In other words, for the case of the WDNR dielectric, each absorption peak acts as two independent single resonances for the precursors but acts as a combined resonance for the main signal. Furthermore, using the theory, the experimental data are categorized by either the single- or double-resonance models by measuring the size of the fixed-frequency $\Delta_c/2\pi$, the excited state splitting 58 MHz

as the origin of the modulation. The modulation pattern may therefore be used to probe a single- or double-resonance Lorentz dielectric in the WDNR regime.

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